

CS201 Composite

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Abstract—Write only ONE function per question in Java. Do not check the input, assume that the input is correct. Unless noted, (i) the input data structure is not empty, (ii) time complexity of the algorithm does not matter, (iii) you can not use extra data structures, (iv) you can use any function given in the data structure.

I. QUESTION (ALGORITHM ANALYSIS), 20 POINTS

You are given the following recurrence equation. Solve this without the master theorem.

$$T(N) = 8T(N/2) + O(N^2) \quad (1)$$

II. QUESTION (LINKED LIST), 15 POINTS

Write the method which removes the nodes that appear in the list2 from the original list. You are not allowed to use any methods from the LinkedList class. You can assume both the original list and list2 do not contain duplicate elements. Do not modify list2. Your method should run in $\mathcal{O}(N^2)$ time.

```
public void remove(LinkedList list2)
```

III. QUESTION (TREE), 15 POINTS

Write the recursive method in TreeNode class, which collects all values in all nodes in the tree in a sorted manner. You are not allowed to use any tree methods.

```
int [] collectNodes()
```

IV. QUESTION (GRAPH), 15 POINTS

For a directed weighted graph, write the method in linked list implementation which returns the shortest distance between the nodes index1 and index2 by two hops, that is, it will return of the shortest of all paths, where one goes from index1 node to node i , then from node i to node $index2$.

```
int shortestIn2Hops(int index1, int index2)
```

V. QUESTION (SORTING), 15 POINTS

Suppose arrays representing sets A and B are both sorted. Write a linear time method that finds if A is a subset of B.

```
boolean isSubset(int [] A, int [] B)
```

VI. QUESTION (SORTING), 20 POINTS

Given an array A of N integers, write a function. Your function should display the three most frequent elements in the given array A, which is already sorted. Your function must run in $\mathcal{O}(N)$ time. You should not modify the input array A.

```
int [] topThreeFrequent(int [] A)
```